

WARING DECOMPOSITION OF DERIVATIVES AND ITS APPLICATIONS

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Abstract. Computing high-order derivatives in neural network residuals is often computationally and memory intensive, primarily due to the large computational graphs caused by automatic differentiation. In this work, we introduce the Waring decomposition of derivatives (WDD) method, which expresses a derivative as a weighted sum of directional Taylor coefficients. We prove that the minimal number of complex directional evaluations required to recover a target partial derivative equals the Waring rank of its corresponding monomial, and a classical roots-of-unity Waring decomposition attains this minimum. Experiments on individual high-order partial derivatives assess the runtime of the WDD method relative to nested coordinate Jacobian–vector products. Numerical experiments show that the WDD method reduces derivative-evaluation time. In PINN training for the KdV and Cahn–Hilliard equations, it completes more updates and attains lower mean solution errors than nested coordinate JVPs under a common training-time budget, with both interior and constraint training points resampled at every update.

Key words. high-order derivatives, Taylor-mode automatic differentiation, Waring decomposition, directional Taylor coefficients

AMS subject classifications. 65D25, 14N07

1. Introduction. Physics-informed neural networks approximate solutions of differential equations by minimizing loss functions constructed from the corresponding PDE residuals. This requires derivatives of the network with respect to its inputs. For high-order equations, the repeated evaluation of these derivatives can dominate both the runtime and memory cost of residual evaluation.

Automatic differentiation is the standard tool for evaluating these derivatives [1, 9, 20]. Several exact approaches reduce its cost by exploiting the structure of the differential operator. For example, forward Laplacian methods propagate the Laplacian through the network without constructing the full Hessian [18]. Such methods can be efficient, but their differentiation rules are designed for specific structured operators and do not readily extend to general high-order partial derivatives. Taylor-mode automatic differentiation provides a more general alternative by propagating a truncated univariate Taylor series through the network [2, 9]. Existing general constructions recover multivariate high-order derivatives from multiple Taylor expansions, often by reconstructing the full derivative tensor [8]. The number of tensor components grows rapidly with both the input dimension and the derivative order, resulting in substantial computational and GPU-memory costs [23].

Randomization offers another route to reducing derivative costs. Stochastic dimension gradient descent uses coordinate sampling to evaluate only a subset of derivative terms during each optimization step [13]. Hutchinson-type estimators use random projections to approximate trace-type differential operators [11, 14, 21]. However, both approaches still rely on automatic differentiation to compute the sampled derivatives or projected vector products. A separate line of work avoids automatic differentiation entirely when computing derivatives. Finite differences approximate derivatives from nearby network evaluations [6, 7], whereas randomized smoothing uses Stein’s identity to express derivatives of a smoothed network as expectations of perturbed function values [10, 27]. STDE takes a different route by combining randomization

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with high-order automatic differentiation: it constructs randomized input tangents so that general linear differential operators can be estimated through univariate Taylor-mode propagation [23]. STDE++ also discusses the exact recovery of mixed partial derivatives from linear combinations of directional Taylor coefficients. It further develops STDE by providing polynomial-time Taylor-mode constructions, improving on earlier general schemes whose cost grows exponentially with the derivative order [24]. Although these methods reduce computational costs, their performance depends on the choice of method-specific parameters. Finite differences require a step size that balances truncation and round-off errors, randomized smoothing depends on both the smoothing bandwidth and the Monte Carlo sample size, and the randomized estimators require a sampling distribution and budget. For the stochastic methods, these choices directly affect estimator variance: insufficient or poorly matched sampling can yield high-variance residual estimates, while a small smoothing bandwidth can sharply amplify variance for high-order derivatives and may destabilize PINN training [10, 12, 24].

In this paper, we introduce the Waring decomposition of derivatives (WDD) method for evaluating homogeneous constant-coefficient differential operators through directional Taylor coefficients, and specialize it to mixed partial derivatives. Our main contributions are:

- We establish an exact correspondence between directional Taylor representations of homogeneous constant-coefficient differential operators and power-sum identities for their polynomial symbols. This correspondence identifies the minimum number of complex directions with the Waring rank of the symbol.
- For a single mixed partial derivative, we combine the correspondence with the established complex Waring-rank and roots-of-unity results for monomials to obtain a closed-form representation with the minimum number of complex directions. A batched Taylor-mode implementation evaluates the resulting directional Taylor coefficients independently and in parallel.
- We compare the WDD method with nested coordinate Jacobian–vector products (JVPs) on individual high-order partial derivatives, examining runtime across derivative patterns and input batch sizes without outer whole-function JIT compilation. KdV and Cahn–Hilliard PINN comparisons measure the number of complete updates and the solution error under a common wall-time budget, with matched initialization and resampled interior and constraint-point sequences.

The remainder of the paper is organized as follows. Section 2 introduces the method of PINNs, multi-index notation, and directional Taylor coefficients. Section 3 establishes the operator–symbol correspondence, analyzes the computational complexity, specializes the correspondence to monomial Waring decompositions, and presents the resulting minimum-direction formula with its batched Taylor realization. Section 4 evaluates the derivative backend on individual high-order partial derivatives and examines its use in complete KdV and Cahn–Hilliard PINN training. Section 5 concludes. A coefficient verification of the cited roots-of-unity decomposition and several representative schedules are collected in Appendix A.

2. Preliminaries.

2.1. Method of PINNs. Let $\Omega \subset \mathbb{R}^d$ be a computational domain and $\Gamma \subseteq \partial\Omega$ the part of its boundary on which constraints are prescribed. For evolution equations, Ω is the space–time domain, d includes the time coordinate, and Γ includes the initial surface and the prescribed spatial boundary surfaces. We consider the problem

$$\mathcal{P}u_\star = f \quad \text{in } \Omega, \quad \mathcal{B}u_\star = g \quad \text{on } \Gamma, \quad (1)$$

where $\mathcal{P}$ is a differential operator and $\mathcal{B}$ is a boundary operator or a collection of boundary operators. In the PINN framework, $u_\star$ is approximated by a neural network u_θ [15, 17, 22, 26]. Let $(\mathbf{x}_{r,i})_{i=1}^{n_r} \subset \Omega$ and $(\mathbf{x}_{b,i})_{i=1}^{n_b} \subset \Gamma$ denote the residual and constraint training points. For weights $\lambda_r, \lambda_b > 0$, the network parameters are obtained by minimizing

$$\mathcal{J}(\theta) := \frac{\lambda_r}{n_r} \sum_{i=1}^{n_r} |\mathcal{P}u_\theta(\mathbf{x}_{r,i}) - f(\mathbf{x}_{r,i})|^2 + \frac{\lambda_b}{n_b} \sum_{i=1}^{n_b} \|\mathcal{B}u_\theta(\mathbf{x}_{b,i}) - g(\mathbf{x}_{b,i})\|^2. \quad (2)$$

Evaluating the PDE residual in (2) requires derivatives of u_θ . For high-order equations, nested automatic differentiation can dominate the time and memory required for residual evaluation because it retains large intermediate computational graphs and repeatedly differentiates through the network.

2.2. Multi-index notation. Let $\mathbb{N}_0$ denote the set of nonnegative integers. For each coordinate x_i , let ∂_i denote differentiation with respect to x_i . For a multi-index $\beta = (\beta_1, \dots, \beta_d) \in \mathbb{N}_0^d$, write

$$|\beta| := \sum_{i=1}^d \beta_i, \quad \beta! := \prod_{i=1}^d \beta_i!, \quad \partial^\beta := \partial_1^{\beta_1} \cdots \partial_d^{\beta_d}. \quad (3)$$

For $\mathbf{v} = (v_1, \dots, v_d) \in \mathbb{C}^d$, we also write $\mathbf{v}^\beta := v_1^{\beta_1} \cdots v_d^{\beta_d}$. The zero multi-index is denoted by $\mathbf{0}$, so that $\partial^{\mathbf{0}}u = u$.

2.3. Directional Taylor coefficients. At each evaluation point $\mathbf{x}$, assume that u admits a holomorphic extension to an open neighborhood of $\mathbf{x}$ in $\mathbb{C}^d$, again denoted by u . In the PINN setting, $u = u_\theta$. For every integer $k \geq 1$ and every $\mathbf{v} \in \mathbb{C}^d$, define the order- k directional Taylor coefficient by

$$T_k(\mathbf{x}; \mathbf{v}) := \frac{1}{k!} \frac{d^k}{d\tau^k} u(\mathbf{x} + \tau \mathbf{v}) \Big|_{\tau=0}. \quad (4)$$

Thus, $T_k(\mathbf{x}; \mathbf{v})$ is the coefficient of τ^k in the Taylor expansion of $u(\mathbf{x} + \tau \mathbf{v})$; we set $T_0(\mathbf{x}; \mathbf{v}) = u(\mathbf{x})$. The chain rule gives

$$T_k(\mathbf{x}; \mathbf{v}) = \frac{1}{k!} D^k u(\mathbf{x})[\mathbf{v}, \dots, \mathbf{v}], \quad (5)$$

where $D^k u(\mathbf{x})$ denotes the symmetric complex k -linear derivative of u at $\mathbf{x}$.

Writing $\mathbf{v} = (v_1, \dots, v_d)$, the multivariate Taylor formula [19, Sec. 3.17] gives

$$T_k(\mathbf{x}; \mathbf{v}) = \sum_{|\beta|=k} \frac{\mathbf{v}^\beta}{\beta!} \partial^\beta u(\mathbf{x}). \quad (6)$$

3. Polynomial symbols and directional representations. In this section, we first establish an exact correspondence between directional Taylor representations of homogeneous constant-coefficient differential operators and power-sum identities for their polynomial symbols. We next derive a general direction bound and compare the computational complexity with nested coordinate JVPs. We then specialize the correspondence to a mixed partial derivative, whose monomial symbol permits a minimum-direction formula through established Waring-rank results.

3.1. Exact operator–symbol correspondence. Let

$$\mathcal{L} = \sum_{|\beta|=p} a_\beta \partial^\beta, \quad a_\beta \in \mathbb{C}, \quad p \geq 1, \quad (7)$$

be a nonzero homogeneous constant-coefficient differential operator. We seek to express $\mathcal{L}u$ as a linear combination of directional Taylor coefficients, using their efficient evaluation to accelerate the computation. We first establish the following proposition.

PROPOSITION 3.1 (Exact operator–symbol correspondence). *Let R be a positive integer, and let $c_r \in \mathbb{C}$ and $\mathbf{v}_r \in \mathbb{C}^d$ for $r = 1, \dots, R$. For formal polynomial variables $\mathbf{z} = (z_1, \dots, z_d)$, we denote the standard bilinear pairing by $\mathbf{v} \cdot \mathbf{z} := \sum_{j=1}^d v_j z_j$. Then the identity*

$$\mathcal{L}u(\mathbf{x}) = \sum_{r=1}^R c_r T_p(\mathbf{x}; \mathbf{v}_r) \quad (8)$$

holds for every u satisfying the assumption in Section 2.3 and every evaluation point $\mathbf{x}$ if and only if

$$\sum_{|\beta|=p} a_\beta \mathbf{z}^\beta = \frac{1}{p!} \sum_{r=1}^R c_r (\mathbf{v}_r \cdot \mathbf{z})^p \quad \text{in } \mathbb{C}[z_1, \dots, z_d]. \quad (9)$$

Proof. We show that both (8) and (9) are equivalent to

$$\sum_{r=1}^R c_r \mathbf{v}_r^\beta = \beta! a_\beta, \quad \text{for every } \beta \in \mathbb{N}_0^d \text{ with } |\beta| = p. \quad (10)$$

First, substituting (6) into the right-hand side of (8) gives

$$\sum_{r=1}^R c_r T_p(\mathbf{x}; \mathbf{v}_r) = \sum_{|\beta|=p} \left(\frac{1}{\beta!} \sum_{r=1}^R c_r \mathbf{v}_r^\beta \right) \partial^\beta u(\mathbf{x}). \quad (11)$$

Hence, (10) implies (8). Conversely, for any β with $|\beta| = p$, choose $u_\beta(\mathbf{y}) = (\mathbf{y} - \mathbf{x})^\beta$. Substituting u_β into (8) gives

$$\beta! a_\beta = \sum_{r=1}^R c_r \mathbf{v}_r^\beta.$$

Therefore, (8) and (10) are equivalent.

On the other hand, the multinomial theorem gives

$$\frac{1}{p!} \sum_{r=1}^R c_r (\mathbf{v}_r \cdot \mathbf{z})^p = \sum_{|\beta|=p} \left(\frac{1}{\beta!} \sum_{r=1}^R c_r \mathbf{v}_r^\beta \right) \mathbf{z}^\beta.$$

Comparing polynomial coefficients in (9) shows that (9) is equivalent to (10). $\square$

Proposition 3.1 shows that evaluating $\mathcal{L}u(\mathbf{x})$ reduces to evaluating a linear combination of directional Taylor coefficients. This gives the general WDD procedure in Algorithm 1.

Algorithm 1 WDD evaluation of $\mathcal{L}u(\mathbf{x})$ with $\mathcal{L}$ given in (7)

Input: Function u satisfying the assumption in Section 2.3; evaluation point $\mathbf{x} \in \Omega$; homogeneous operator $\mathcal{L}$ of order p given in (7); coefficients $c_1, \dots, c_R$ and directions $\mathbf{v}_1, \dots, \mathbf{v}_R$ satisfying (9).

Output: The value of $\mathcal{L}u(\mathbf{x})$.

1. For $r = 1, 2, \dots, R$, compute $T_p(\mathbf{x}; \mathbf{v}_r)$ by Taylor-mode automatic differentiation.
 2. Compute $\mathcal{L}u(\mathbf{x}) = \sum_{r=1}^R c_r T_p(\mathbf{x}; \mathbf{v}_r)$.
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Algorithm 1 does not specify how to obtain the power-sum representation in (9). Its existence is discussed in Section 3.2, while the minimum number of directions is related to the following concept of Waring rank.

DEFINITION 3.1 (Complex Waring decomposition and rank). *Let $P \in \mathbb{C}[z_1, \dots, z_d]$ be a nonzero homogeneous polynomial of degree $p \geq 1$. A complex Waring decomposition of P of length R is an identity*

$$P(\mathbf{z}) = \sum_{r=1}^R \lambda_r \ell_r(\mathbf{z})^p,$$

where R is a positive integer, $\lambda_r \in \mathbb{C} \setminus \{0\}$, and each ℓ_r is a nonzero complex linear form. The complex Waring rank $R_{\mathbb{C}}(P)$ is the smallest possible length of such a decomposition [5, Section 1].

By Proposition 3.1, the minimum number of directions in (8) is $R_{\mathbb{C}}\left(\sum_{|\beta|=p} a_{\beta} \mathbf{z}^{\beta}\right)$. For a homogeneous polynomial with rational coefficients, determining the complex Waring rank is NP-hard [25]. Nevertheless, a polynomial basis provides a direction bound that allows us to analyse the computational cost without requiring a minimum decomposition.

3.2. Computational complexity. Algorithm 1 relies on the existence of a valid power-sum representation (9). We first resolve this existence question for an arbitrary homogeneous constant-coefficient operator and provide a general upper bound on the number of required directions R .

PROPOSITION 3.2 (Existence of a directional representation). *For every operator $\mathcal{L}$ in (7), there exist a positive integer R , coefficients $c_1, \dots, c_R \in \mathbb{C}$, and directions $\mathbf{v}_1, \dots, \mathbf{v}_R \in \mathbb{C}^d$ satisfying (9), with*

$$R \leq \binom{d+p-1}{p}. \quad (12)$$

Proof. The space of complex homogeneous polynomials of degree p in d variables has dimension $\binom{d+p-1}{p}$ and admits a basis consisting of $\binom{d+p-1}{p}$ p th powers of linear forms [3, Sec. 3.1]. Expressing the polynomial symbol $\sum_{|\beta|=p} a_{\beta} \mathbf{z}^{\beta}$ on the left-hand side of (9) in this basis yields the required representation with at most $\binom{d+p-1}{p}$ terms. $\square$

Proposition 3.2 guarantees that any such operator $\mathcal{L}$ can be evaluated using at most $\binom{d+p-1}{p}$ directional Taylor coefficients. To examine the computational characteristics of this directional representation, we fix the network architecture and count arithmetic operations per evaluation point. The directions and coefficients supplied by Proposition 3.2 are precomputed.

For comparison, consider the nested coordinate JVP baseline evaluated in Section 4. Computing an order- p derivative by nested forward-mode automatic differentiation repeatedly pairs each propagated component with its directional tangent, yielding 2^p components after p nesting levels [2]. Propagating these components separately produces exponential growth in arithmetic work with p , as also discussed for repeated backward-mode AD [23, Sec. 3.2].

In contrast, for a fixed direction $\mathbf{v}$, evaluating the directional Taylor coefficient $T_p(\mathbf{x}; \mathbf{v})$ propagates univariate Taylor expansions through the network. Under standard Taylor recurrences, evaluating the coefficients of orders $0, \dots, p$ incurs $O(p^2)$ arithmetic operations [23, Sec. 4.2]. Evaluating all R directions and their weighted sum therefore costs $O(Rp^2)$. Moreover, with the forward trace retained, reverse-mode differentiation of a scalar loss with respect to the network parameters θ adds at most a constant multiple of the forward work [1, Sec. 3.2]. Thus, the combined forward and parameter-backpropagation cost of the general WDD procedure in Algorithm 1 is

$$C_{\text{WDD}} = O(Rp^2). \quad (13)$$

Substituting (12) into (13) yields

$$C_{\text{WDD}} = O\left(p^2 \binom{d+p-1}{p}\right). \quad (14)$$

For any fixed input dimension d , the binomial factor $\binom{d+p-1}{p} = O(p^{d-1})$ grows polynomially with the derivative order p , avoiding the exponential scaling in component propagation of nested coordinate JVPs. In addition, the directional Taylor evaluations across the R directions are mutually independent and can be batched together.

For common differential operators, the number of required directions is typically smaller than the upper bound in (12). When $\mathcal{L}$ is a specific monomial partial derivative, the minimal number of directions R can be much smaller than $\binom{d+p-1}{p}$. In the next subsection, we specialize the construction to monomial symbols and determine the exact minimum direction count via the complex Waring rank.

3.3. Monomial specialization and complex Waring rank. Fix a nonzero multi-index $\nu \in \mathbb{N}_0^d$ of order $p := |\nu| \geq 1$, and consider the partial derivative operator $\mathcal{L} = \partial^\nu$. Applying Proposition 3.1 gives

$$\partial^\nu u(\mathbf{x}) = \sum_{r=1}^R c_r T_p(\mathbf{x}; \mathbf{v}_r) \iff p! \mathbf{z}^\nu = \sum_{r=1}^R c_r (\mathbf{v}_r \cdot \mathbf{z})^p. \quad (15)$$

To state the rank formula and an attaining decomposition, relabel the variables with positive exponents in $\mathbf{z}^\nu$ after a permutation of the coordinates so that

$$\mathbf{z}^\nu = z_0^{\nu_0} \cdots z_n^{\nu_n}, \quad 1 \leq \nu_0 \leq \cdots \leq \nu_n, \quad p = \sum_{j=0}^n \nu_j. \quad (16)$$

Thus, z_0 is chosen as a variable with the smallest positive exponent. The rank remains unchanged when the linear forms are allowed to involve the remaining polynomial variables [5, Remark 2.3].

For $m \geq 1$, let $U_m := \{\zeta \in \mathbb{C} : \zeta^m = 1\}$, and define

$$\mathcal{U}_\nu := U_{\nu_1+1} \times \cdots \times U_{\nu_n+1}. \quad (17)$$

When $n = 0$, $\mathcal{U}_\nu$ consists of the empty tuple and has cardinality one.

The monomial rank formula is due to Carlini, Catalisano, and Geramita [5, Cor. 3.3 and Remark 2.3], while the explicit roots-of-unity decomposition is given by Buczyńska, Buczyński, and Teitler [4, Sec. 2, Eq. (8)]. The following theorem records these results in the normalization required by (15).

THEOREM 3.3 (Complex Waring rank and roots-of-unity decomposition of a monomial). *Let the variables and positive exponents of $\mathbf{z}^\nu$ be ordered as in (16). Then every R -term complex Waring decomposition of $p! \mathbf{z}^\nu$ satisfies*

$$R \geq R_{\mathbb{C}}(\mathbf{z}^\nu) = \prod_{j=1}^n (\nu_j + 1). \quad (18)$$

Moreover, the roots-of-unity identity

$$p! \mathbf{z}^\nu = \frac{\nu!}{\prod_{j=1}^n (\nu_j + 1)} \sum_{\zeta \in \mathcal{U}_\nu} \left(\prod_{j=1}^n \zeta_j \right) \left(z_0 + \sum_{j=1}^n \zeta_j z_j \right)^p \quad (19)$$

contains exactly $\prod_{j=1}^n (\nu_j + 1)$ terms and therefore attains the lower bound in (18).

For completeness, Appendix A.2 verifies (19) directly in the notation used here.

3.4. Minimum-direction formula and algorithm. Combining Proposition 3.1 with Theorem 3.3 gives the corresponding minimum-direction representation of a mixed partial.

COROLLARY 3.4 (Minimum-direction representation of a mixed partial). *Let the coordinates and positive entries of ν be ordered as in (16). Then*

$$\partial^\nu u(\mathbf{x}) = \frac{\nu!}{\prod_{j=1}^n (\nu_j + 1)} \sum_{\zeta \in \mathcal{U}_\nu} \left(\prod_{j=1}^n \zeta_j \right) T_p(\mathbf{x}; (1, \zeta_1, \dots, \zeta_n, 0, \dots, 0)). \quad (20)$$

Moreover, among all coefficients $c_1, \dots, c_R \in \mathbb{C}$ and directions $\mathbf{v}_1, \dots, \mathbf{v}_R \in \mathbb{C}^d$ for which the left-hand identity in (15) holds for every u satisfying the assumption in Section 2.3, the minimum possible value of R is

$$\prod_{j=1}^n (\nu_j + 1). \quad (21)$$

Proof. Applying Proposition 3.1 to (19) gives (20). Conversely, the equivalence in (15) maps every such R -term representation, after omitting zero terms, to a Waring decomposition of $p! \mathbf{z}^\nu$ with at most R terms. The lower bound (18) gives the stated minimum, which is attained by (20). $\square$

The rank formula also describes when the directional reduction is most effective. A pure derivative requires one direction, whereas a square-free derivative of order p

requires 2^{p-1} directions; repeated indices can reduce this number. The minimum in Corollary 3.4 concerns one fixed monomial derivative and exact linear representations by complex order- p directional Taylor coefficients.

Algorithm 2 gives the specialized WDD procedure for evaluating the roots-of-unity formula in Corollary 3.4.

Algorithm 2 WDD evaluation of $\partial^\nu u(\mathbf{x})$ using the roots-of-unity formula (20)

Input: Function u satisfying the assumption in Section 2.3; evaluation point $\mathbf{x} \in \Omega$; multi-index $\nu \in \mathbb{N}_0^d$ with $p = |\nu| \geq 1$.

Output: The value of $\partial^\nu u(\mathbf{x})$.

1. Order the coordinates associated with the positive entries of ν as in (16), retaining their positions in the original coordinate system.
 2. Form the roots-of-unity index set $\mathcal{U}_\nu$ defined in (17).
 3. For each $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathcal{U}_\nu$, construct $\mathbf{v}_\zeta \in \mathbb{C}^d$ with entries $(1, \zeta_1, \dots, \zeta_n)$ in these ordered coordinate positions and zero entries in all remaining positions.
 4. Compute $T_p(\mathbf{x}; \mathbf{v}_\zeta)$ for each $\zeta \in \mathcal{U}_\nu$ by Taylor-mode automatic differentiation.
 5. Evaluate the weighted sum in (20) to obtain $\partial^\nu u(\mathbf{x})$.
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4. Numerical experiments. We evaluate the proposed Waring decomposition of derivatives (WDD) method in two complementary settings. First, we measure the evaluation time of individual high-order partial derivatives across various orders and multi-index patterns, using the minimum-direction complex Waring representation in Corollary 3.4. Second, we assess PINN training on three nonlinear differential equations: a one-dimensional KdV equation with a gradient-enhanced objective, a two-dimensional KdV equation with mixed derivatives, and a fourth-order Cahn–Hilliard equation. We apply Algorithm 1, constructing directional Taylor representations that facilitate coefficient reuse across different derivative terms. In every experimental setting, the WDD method and the nested coordinate Jacobian–vector product (JVP) baseline compute the same mathematical derivatives; in the PDE experiments, they also minimize identical loss objectives under matched computational conditions.

4.1. Individual high-order partial derivatives. We evaluate the evaluation time of individual high-order partial derivatives $u_{i_1 \dots i_p} = \partial_{i_1} \dots \partial_{i_p} u$ across derivative orders p and multi-index patterns at batch size $B = 100$. The test network has 16 inputs and three width-128 tanh hidden layers, evaluated on identical weights and inputs for both methods. Table 4.1 reports the speedup rate relative to nested JVP of the six individual derivatives.

TABLE 4.1
Evaluation times and speedups for individual high-order partial derivatives.

Target	p	R	B	Nested JVP (ms)	WDD (ms)	Speedup
u_{111111}	6	1	100	612.407 ± 0.865	4.416 ± 0.029	$138.67\times$
u_{111222}	6	4	100	617.441 ± 0.719	4.736 ± 0.127	$130.42\times$
u_{112233}	6	9	100	624.880 ± 1.970	4.691 ± 0.030	$133.21\times$
u_{123456}	6	32	100	617.041 ± 0.759	5.948 ± 0.129	$103.78\times$
$u_{11223344}$	8	27	100	5138.625 ± 4.745	21.734 ± 0.004	$236.43\times$
$u_{1112223333}$	9	16	100	15653.074 ± 3.763	13.802 ± 0.009	$1134.13\times$

4.2. PINN training on nonlinear differential equations. For the PDE tests, u_θ has four fully connected hidden layers of width 128, tanh activations, and

a scalar linear output. The inputs are the physical coordinates. Adam [16] uses a constant learning rate of 10^{-5} for both KdV problems and 10^{-4} for Cahn–Hilliard. Parameter gradients are computed by reverse-mode differentiation. Each update resamples 400 interior space–time points and independently resamples points on each prescribed initial and spatial boundary face, all uniformly: 128 points per face for the one-dimensional problem and 256 for the two-dimensional problems. Conditions on the same face use the same points. Interior and constraint points use independent random streams. Across five random seeds, paired methods share the initialization, loss weights, and samples at each common update index.

In the objectives below, $\langle \cdot \rangle_r$ denotes the mean over interior points. Each PDE example specifies its initial and boundary penalty $\mathcal{J}_b$.

Each PDE run is trained for 1200 seconds. Timing covers sampling, derivative evaluation, parameter backpropagation, optimizer updates, finite-value checks of the loss and gradients, logging, and periodic checkpoint writes. Before timing, one forward and backward pass is used for warm-up without updating parameters. Initial and terminal checkpoint writes and post-training diagnostics are excluded; an update in progress at the deadline is completed. For each paired seed, the completed-update ratio S_{iter} is the number of completed Taylor updates divided by the number of completed nested JVP updates.

For a fixed evaluation set D , the relative solution error is

$$\text{RE} := \frac{(\sum_{\mathbf{x} \in D} |u_\theta(\mathbf{x}) - u_\star(\mathbf{x})|^2)^{1/2}}{(\sum_{\mathbf{x} \in D} |u_\star(\mathbf{x})|^2)^{1/2}}. \quad (22)$$

We evaluate RE on 10,000 uniform space–time points and, separately, on 10,000 uniform spatial points at the terminal time $t = 1$. These test sets are fixed, independent of training, and shared by the paired methods. The PDE tables report the mean and sample standard deviation of update counts, final-iterate errors, and paired completed-update ratios. The figures plot space–time relative error against elapsed training time. Saved models are evaluated after training; their log errors are interpolated onto a common time grid without smoothing or extrapolation. Solid curves show the arithmetic mean across seeds, and shading extends one sample standard deviation upward.

4.2.1. Gradient-enhanced PINN for the KdV equation. This example tests the evaluation of fourth-order derivatives arising from a gradient-enhanced PINN objective. We solve

$$\begin{cases} u_t + uu_x + 0.0025u_{xxx} = 0, & x \in (-1, 1), \quad t \in (0, 1), \\ u(x, 0) = u_\star(x, 0), & x \in [-1, 1], \\ u(-1, t) = u_\star(-1, t), & t \in (0, 1), \\ u(1, t) = u_\star(1, t), \quad \partial_x u(1, t) = \partial_x u_\star(1, t), & t \in (0, 1). \end{cases} \quad (23)$$

The exact solution is the traveling solitary wave

$$u_\star(x, t) = 0.75 \operatorname{sech}^2(5(x + 0.5 - 0.25t)). \quad (24)$$

No interior solution values enter the training loss. Define the PDE residual by

$$r_\theta(x, t) := (u_\theta)_t + u_\theta(u_\theta)_x + 0.0025(u_\theta)_{xxx},$$

and the initial and boundary penalty by

$$\begin{aligned}\mathcal{J}_b(\theta) = & \text{MSE}_x[u_\theta(x, 0) - u_\star(x, 0)] \\ & + \text{MSE}_t[u_\theta(-1, t) - u_\star(-1, t)] \\ & + \text{MSE}_t[u_\theta(1, t) - u_\star(1, t)] \\ & + \text{MSE}_t[\partial_x u_\theta(1, t) - \partial_x u_\star(1, t)],\end{aligned}$$

where MSE_x and MSE_t denote the sample mean squared errors over $x \in [-1, 1]$ and $t \in (0, 1)$, respectively. The gradient-enhanced objective is

$$\mathcal{J}_{\text{ID}}(\theta) = \langle r_\theta^2 \rangle_r + 10^{-3} [\langle (\partial_x r_\theta)^2 \rangle_r + \langle (\partial_t r_\theta)^2 \rangle_r] + 10\mathcal{J}_b(\theta). \quad (25)$$

In the polynomial identities below, x, y, t also serve as the corresponding formal variables, and we abbreviate $T_k(\mathbf{x}; \mathbf{v})$ by $T_k(\mathbf{v})$. The highest-order derivatives in the residual gradients have symbols x^4 and x^3t . The latter satisfies

$$x^3t = \frac{(x+t)^4 - (x-t)^4}{6} - \frac{(x+2t)^4 - (x-2t)^4}{48}. \quad (26)$$

Applying Proposition 3.1 to (26) and x^4 gives

$$\begin{aligned}u_{xxxx} &= 24T_4(1, 0), \\ u_{xxxt} &= 4[T_4(1, 1) - T_4(1, -1)] - \frac{1}{2}[T_4(1, 2) - T_4(1, -2)].\end{aligned} \quad (27)$$

Algorithm 1 evaluates Taylor expansions along five real directions. Their lower-order coefficients also supply the remaining derivatives in (25).

Table 4.2 and Figure 4.1 show that the WDD method completes more updates and attains lower mean final errors within the same training-time budget.

TABLE 4.2
Numerical results for the one-dimensional KdV equation (23).

Method	Updates	Space-time RE	Terminal RE	S_{iter}
Nested JVP	2928.8 ± 63.7	$5.365 \times 10^{-1} \pm 3.105 \times 10^{-2}$	$5.950 \times 10^{-1} \pm 2.053 \times 10^{-2}$	1
WDD	22164.4 ± 686.6	$6.783 \times 10^{-3} \pm 1.490 \times 10^{-3}$	$1.185 \times 10^{-2} \pm 2.697 \times 10^{-3}$	7.57 ± 0.26

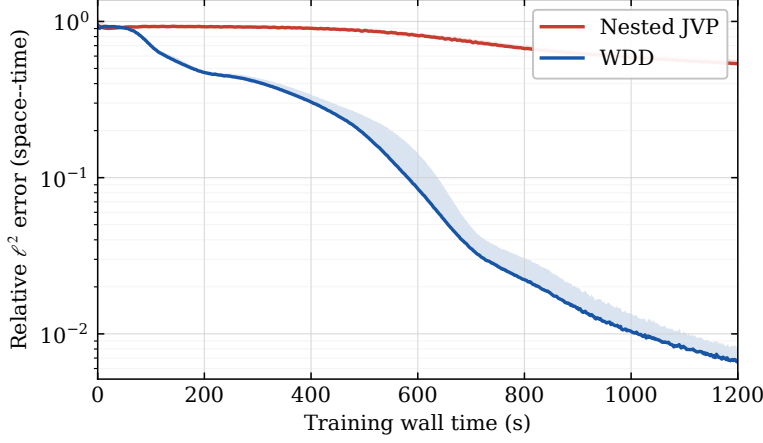


FIG. 4.1. Space-time relative error during training for the one-dimensional KdV equation (23).

4.2.2. Mixed-derivative PINN for a two-dimensional KdV equation.

This example combines the spatial mixed derivative u_{xxxx} with the space-time derivative u_{ty} . We consider the KdV-type equation in [23, Eq. (20)],

$$\begin{cases} u_{ty} + u_{xxxx} + 3(u_x u_y)_x - u_{xx} + 2u_{yy} = 0, & (x, y) \in (-1, 1)^2, \quad t \in (0, 1), \\ u(x, y, 0) = u_*(x, y, 0), & (x, y) \in [-1, 1]^2, \\ u(-1, y, t) = u_*(-1, y, t), & y \in [-1, 1], \quad t \in (0, 1), \\ u(1, y, t) = u_*(1, y, t), & y \in [-1, 1], \quad t \in (0, 1), \\ \partial_x u(1, y, t) = \partial_x u_*(1, y, t), & y \in [-1, 1], \quad t \in (0, 1), \\ u(x, -1, t) = u_*(x, -1, t), & x \in [-1, 1], \quad t \in (0, 1), \\ \partial_y u(x, -1, t) = \partial_y u_*(x, -1, t), & x \in [-1, 1], \quad t \in (0, 1). \end{cases} \quad (28)$$

The exact oblique-kink solution is

$$u_*(x, y, t) = \tanh((x + y)/2 - t). \quad (29)$$

Only the PDE is taken from the cited work; the domain, exact solution, and initial-boundary conditions specify the present benchmark. Define the PDE residual by

$$r_\theta(x, y, t) := (u_\theta)_{ty} + (u_\theta)_{xxxx} + 3\partial_x[(u_\theta)_x(u_\theta)_y] - (u_\theta)_{xx} + 2(u_\theta)_{yy},$$

and the initial and boundary penalty by

$$\begin{aligned} \mathcal{J}_b(\theta) = & \text{MSE}_{x,y}[u_\theta(x, y, 0) - u_*(x, y, 0)] \\ & + \text{MSE}_{y,t}[u_\theta(-1, y, t) - u_*(-1, y, t)] \\ & + \text{MSE}_{y,t}[u_\theta(1, y, t) - u_*(1, y, t)] \\ & + \text{MSE}_{x,t}[u_\theta(x, -1, t) - u_*(x, -1, t)] \\ & + \text{MSE}_{y,t}[\partial_x u_\theta(1, y, t) - \partial_x u_*(1, y, t)] \\ & + \text{MSE}_{x,t}[\partial_y u_\theta(x, -1, t) - \partial_y u_*(x, -1, t)], \end{aligned}$$

where the subscripts of MSE indicate the coordinates over which the sample mean squared errors are taken. The objective is

$$\mathcal{J}_{2D}(\theta) = \langle r_\theta^2 \rangle_r + 10\mathcal{J}_b(\theta). \quad (30)$$

There is no residual-gradient penalty.

Using (26) with t replaced by y , together with

$$yt = \frac{1}{4}[(y+t)^2 - (y-t)^2], \quad (31)$$

Proposition 3.1 gives

$$\begin{aligned} u_{xxxxy} &= 4[T_4(1, 1, 0) - T_4(1, -1, 0)] - \frac{1}{2}[T_4(1, 2, 0) - T_4(1, -2, 0)], \\ u_{ty} &= \frac{1}{2}[T_2(0, 1, 1) - T_2(0, 1, -1)]. \end{aligned} \quad (32)$$

Algorithm 1 evaluates Taylor expansions along six real directions. Their lower-order coefficients also recover $u_x, u_y, u_{xx}, u_{xy}, u_{yy}$.

The results in Table 4.3 and Figure 4.2 show that the computational advantage of the WDD method persists for this mixed-derivative objective.

TABLE 4.3
Numerical results for the two-dimensional KdV equation (28).

Method	Updates	Space-time RE	Terminal RE	S_{iter}
Nested JVP	2992.0 ± 86.8	$4.395 \times 10^{-2} \pm 9.677 \times 10^{-3}$	$6.125 \times 10^{-2} \pm 1.233 \times 10^{-2}$	1
WDD	16653.6 ± 152.2	$1.365 \times 10^{-3} \pm 3.287 \times 10^{-4}$	$2.313 \times 10^{-3} \pm 6.813 \times 10^{-4}$	5.57 ± 0.18

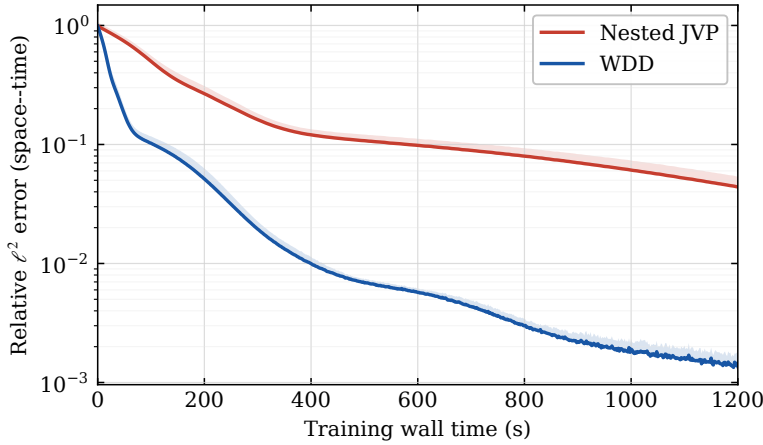


FIG. 4.2. Space-time relative error during training for the two-dimensional KdV equation (28).

4.2.3. Fourth-order Cahn–Hilliard equation. This example tests Taylor coefficient reuse for the Laplacian and biharmonic operators in a nonlinear PDE. We solve the Cahn–Hilliard problem

$$\begin{cases} u_t - \Delta(u^3 - u) + 0.01\Delta^2 u = f(x, y, t), & (x, y) \in (0, \pi)^2, \quad t \in (0, 1), \\ u(x, y, 0) = u_*(x, y, 0), & (x, y) \in [0, \pi]^2, \\ \partial_{\mathbf{n}} u = 0, \quad \partial_{\mathbf{n}} \Delta u = 0, & (x, y) \in \partial((0, \pi)^2), \\ & t \in (0, 1). \end{cases} \quad (33)$$

Here ∇ and Δ act on the spatial coordinates, and $\mathbf{n}$ is the outward spatial unit normal. The exact solution contains three spatial modes,

$$u_\star(x, y, t) = e^{-t} \left[\frac{1}{2} \cos x \cos y + \frac{1}{4} \cos(2x) \cos y + \frac{1}{4} \cos x \cos(2y) \right]. \quad (34)$$

The source f is obtained by substituting (34) into the PDE in (33).

Expanding the nonlinear Laplacian gives the residual

$$r_\theta(x, y, t) := (u_\theta)_t - (3u_\theta^2 - 1)\Delta u_\theta - 6u_\theta |\nabla u_\theta|^2 + 0.01\Delta^2 u_\theta - f. \quad (35)$$

Let $\mathcal{F}$ denote the four spatial boundary faces. The initial and boundary penalty is

$$\begin{aligned} \mathcal{J}_b(\theta) = & \text{MSE}_{x,y} [u_\theta(x, y, 0) - u_\star(x, y, 0)] \\ & + \frac{1}{4} \sum_{F \in \mathcal{F}} \{ \text{MSE}_{F,t} [\partial_{\mathbf{n}} u_\theta] + \text{MSE}_{F,t} [\partial_{\mathbf{n}} \Delta u_\theta] \}, \end{aligned}$$

where $\text{MSE}_{F,t}$ is evaluated over sampled points on $F \times (0, 1)$. The objective is

$$\mathcal{J}_{\text{CH}}(\theta) = \langle r_\theta^2 \rangle_r + 10\mathcal{J}_b(\theta). \quad (36)$$

There is no additional mass, energy, or residual-gradient penalty.

For the spatial operators, we use the identities

$$\begin{aligned} (x^2 + y^2)^2 &= \frac{8}{9} \left[x^4 + \left(\frac{x + \sqrt{3}y}{2} \right)^4 + \left(\frac{-x + \sqrt{3}y}{2} \right)^4 \right], \\ x^2 + y^2 &= \frac{2}{3} \left[x^2 + \left(\frac{x + \sqrt{3}y}{2} \right)^2 + \left(\frac{-x + \sqrt{3}y}{2} \right)^2 \right]. \end{aligned} \quad (37)$$

Writing $\mathbf{v}_r$ for the coefficient vectors of these three linear forms in (x, y, t) , Proposition 3.1 gives

$$\Delta^2 u_\theta = \frac{64}{3} \sum_{r=1}^3 T_4(\mathbf{v}_r), \quad \Delta u_\theta = \frac{4}{3} \sum_{r=1}^3 T_2(\mathbf{v}_r). \quad (38)$$

The quadratic identity also yields $|\nabla u_\theta|^2 = \frac{2}{3} \sum_{r=1}^3 [T_1(\mathbf{v}_r)]^2$. Algorithm 1 therefore evaluates Taylor expansions along three real spatial directions; time and boundary derivatives use coordinate JVPs in both methods.

Table 4.4 and Figure 4.3 show lower mean final errors with the WDD method, although S_{iter} is smaller than in the KdV examples.

TABLE 4.4
Numerical results for the Cahn–Hilliard equation (33).

Method	Updates	Space–time RE	Terminal RE	S_{iter}
Nested JVP	628.4 ± 13.8	$3.032 \times 10^{-1} \pm 2.792 \times 10^{-2}$	$4.956 \times 10^{-1} \pm 7.889 \times 10^{-2}$	1
WDD	1201.0 ± 30.0	$1.601 \times 10^{-1} \pm 3.638 \times 10^{-2}$	$3.301 \times 10^{-1} \pm 7.128 \times 10^{-2}$	1.91 ± 0.07

Across the PDE experiments, the WDD method completes more updates and attains lower mean errors under the same training-time budget, with both interior and constraint training points resampled at every update.

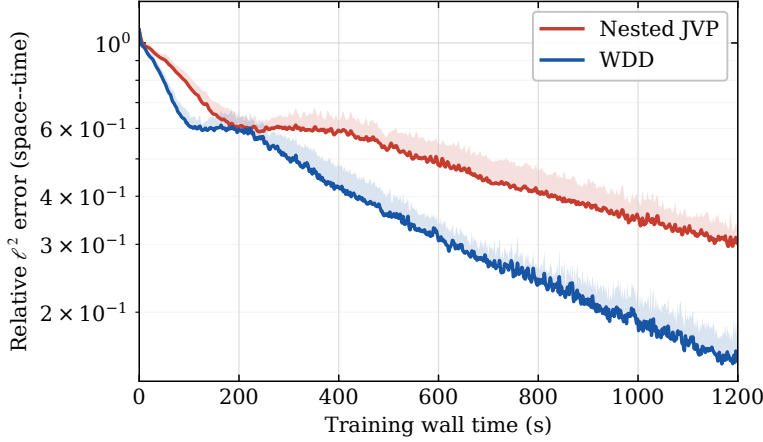


FIG. 4.3. Space-time relative error during training for the Cahn-Hilliard equation (33).

5. Conclusion. Polynomial symbols yield exact directional representations of high-order derivatives. Waring-rank theory and the roots-of-unity decomposition give minimum-length complex formulas for mixed partials. The WDD method evaluates these representations by batched Taylor propagation, while real power-sum identities allow Taylor coefficients to be reused across PDE derivative terms. The experiments show faster individual-partial evaluation and more PINN updates under the same training-time budget. With interior and constraint points resampled at every update, the WDD method attains lower mean errors for the tested KdV and Cahn-Hilliard problems.

Appendix A. Algebraic verification and explicit schedules.

This appendix verifies the cited roots-of-unity decomposition in the notation of this paper and collects several explicit schedules for representative derivative patterns. The formulas in Appendix A.3 are instances of the representation evaluated by the implementation.

A.1. Roots-of-unity filter. LEMMA A.1. *For every integer $m \geq 1$ and every integer k ,*

$$\sum_{\zeta \in U_m} \zeta^k = \begin{cases} m, & m \mid k, \\ 0, & m \nmid k. \end{cases} \quad (39)$$

Proof. Let $\omega := \exp(2\pi i/m)$, so that every element of U_m is ω^ℓ for exactly one $\ell \in \{0, \dots, m-1\}$ and

$$\sum_{\zeta \in U_m} \zeta^k = \sum_{\ell=0}^{m-1} (\omega^\ell)^k.$$

If $m \mid k$, then $\omega^k = 1$ and the sum equals m . If $m \nmid k$, then $\omega^k \neq 1$, and the finite geometric series gives

$$\sum_{\ell=0}^{m-1} (\omega^k)^\ell = \frac{1 - \omega^{km}}{1 - \omega^k} = 0,$$

since $\omega^m = 1$. $\square$

A.2. Verification of the explicit monomial decomposition. For completeness, we verify the decomposition (19) of Buczyńska, Buczyński, and Teitler [4, Sec. 2, Eq. (8)] in the notation used here. For $\beta = (\beta_0, \dots, \beta_n) \in \mathbb{N}_0^{n+1}$ with $|\beta| = p$, the coefficient of $\mathbf{z}^\beta$ on the right-hand side of (19) is

$$\frac{\nu!}{\prod_{j=1}^n (\nu_j + 1)} \binom{p}{\beta} \prod_{j=1}^n \left(\sum_{\zeta_j \in U_{\nu_j+1}} \zeta_j^{\beta_j+1} \right). \quad (40)$$

Lemma A.1 shows that the product can be nonzero only if

$$\beta_j \equiv \nu_j \pmod{\nu_j + 1}, \quad j = 1, \dots, n.$$

Each β_j is therefore either ν_j or at least $2\nu_j + 1$. If $\beta_j \geq 2\nu_j + 1$ for some $j \geq 1$, then the remaining congruences give $\beta_k \geq \nu_k$ and hence

$$\beta_0 = p - \sum_{k=1}^n \beta_k \leq \nu_0 - \nu_j - 1 < 0,$$

which contradicts $\beta_0 \geq 0$. Thus $\beta_j = \nu_j$ for $j = 1, \dots, n$, and $|\beta| = p$ then gives $\beta_0 = \nu_0$. Consequently, the coefficient in (40) vanishes unless $\beta_j = \nu_j$ for every $j = 0, \dots, n$. In that case, the coefficient equals

$$\frac{\nu!}{\prod_{j=1}^n (\nu_j + 1)} \frac{p!}{\nu!} \prod_{j=1}^n (\nu_j + 1) = p!.$$

This proves (19).

A.3. Explicit low-order schedules. The following formulas are low-order instances of (20). Let $\mathbf{e}_i$ denote the i th coordinate vector in $\mathbb{C}^d$.

For the pure derivative ∂_1^p we have $n = 0$, the schedule has one direction, and T_p is evaluated once,

$$\partial_1^p u(\mathbf{x}) = p! T_p(\mathbf{x}; \mathbf{e}_1). \quad (41)$$

For the mixed second derivative, $(\nu_0, \nu_1) = (1, 1)$ and (20) gives

$$\partial_1 \partial_2 u(\mathbf{x}) = \frac{1}{2} T_2(\mathbf{x}; \mathbf{e}_1 + \mathbf{e}_2) - \frac{1}{2} T_2(\mathbf{x}; \mathbf{e}_1 - \mathbf{e}_2). \quad (42)$$

The two directions in (42) attain the minimum in (21).

For the fourth-order mixed derivative $\partial_1^2 \partial_2^2$ we have $n = 1$ and $(\nu_0, \nu_1) = (2, 2)$; with $\omega := \exp(2\pi i/3)$ and $U_3 = \{1, \omega, \omega^2\}$, formula (20) gives the three directions

$$\partial_1^2 \partial_2^2 u(\mathbf{x}) = \frac{4}{3} \sum_{\zeta \in U_3} \zeta T_4(\mathbf{x}; \mathbf{e}_1 + \zeta \mathbf{e}_2). \quad (43)$$

For the sixth-order mixed derivative $\partial_1^2 \partial_2^4$ we have $n = 1$ and $(\nu_0, \nu_1) = (2, 4)$; with U_5 the fifth roots of unity, (20) takes the coordinate of smaller exponent as base and gives the five directions

$$\partial_1^2 \partial_2^4 u(\mathbf{x}) = \frac{48}{5} \sum_{\zeta \in U_5} \zeta T_6(\mathbf{x}; \mathbf{e}_1 + \zeta \mathbf{e}_2). \quad (44)$$

The companion term $\partial_1^4 \partial_2^2$ is the same formula with the two coordinates exchanged. The polynomial identity corresponding to (44) reads

$$6! z_0^2 z_1^4 = \frac{48}{5} \sum_{\zeta \in U_5} \zeta (z_0 + \zeta z_1)^6, \quad (45)$$

and it can be checked directly: the coefficient of $z_0^{\beta_0} z_1^{\beta_1}$ on the right-hand side contains the factor $\sum_{\zeta \in U_5} \zeta^{1+\beta_1}$, which by Lemma A.1 is nonzero only when $\beta_1 \equiv 4 \pmod{5}$. Together with $\beta_0 + \beta_1 = 6$ this forces $\beta = (2, 4)$, and the surviving constant is $\frac{48}{5} \cdot \frac{6!}{2!4!} \cdot 5 = 720 = 6!$. The three-variable pattern $(2, 2, 2)$ of Table A.1 is the same construction with two cube-root factors and nine directions.

Table A.1 lists the minimum number of directions for representative exponent patterns.

TABLE A.1
Minimum numbers of directions for representative monomial derivatives.

pattern	example derivative	$R_{\mathbb{C}}$
(p)	$\partial_1^p u$	1
$(2, 1)$	u_{112}	3
$(1, 1, 1)$	u_{123}	4
$(3, 1)$	u_{1112}	4
$(2, 2)$	u_{1122}	3
$(2, 1, 1)$	u_{1123}	6
$(1, 1, 1, 1)$	u_{1234}	8
$(4, 2)$	u_{111122}	5
$(2, 2, 2)$	u_{112233}	9
(1^6)	u_{123456}	32

REFERENCES

- [1] ATILIM GUNES BAYDIN, BARAK A. PEARLMUTTER, ALEXEY ANDREYEVICH RADUL, AND JEFFREY MARK SISKIND, *Automatic differentiation in machine learning: a survey*, Journal of Machine Learning Research, 18 (2018), pp. 1–43.
- [2] JESSE BETTENCOURT, MATTHEW J. JOHNSON, AND DAVID DUVENAUD, *Taylor-mode automatic differentiation for higher-order derivatives in JAX*, in NeurIPS Workshop on Program Transformations for Machine Learning, 2019.
- [3] GRIGORIY BLEKHERMAN AND ZACH TEITLER, *On maximum, typical and generic ranks*, Mathematische Annalen, 362 (2015), pp. 1021–1031.
- [4] WERONIKA BUCZYŃSKA, JAROSŁAW BUCZYŃSKI, AND ZACH TEITLER, *Waring decompositions of monomials*, Journal of Algebra, 378 (2013), pp. 45–57.
- [5] ENRICO CARLINI, MARIA VIRGINIA CATALISANO, AND ANTHONY V. GERAMITA, *The solution to the Waring problem for monomials and the sum of coprime monomials*, Journal of Algebra, 370 (2012), pp. 5–14.
- [6] PAO-HSIUNG CHIU, JIAN CHENG WONG, CHINCHUN OOI, MY HA DAO, AND YEW-SOON ONG, *CAN-PINN: A fast physics-informed neural network based on coupled-automatic-numerical differentiation method*, Computer Methods in Applied Mechanics and Engineering, 395 (2022), p. 114909.
- [7] BENGT FORNBERG, *Generation of finite difference formulas on arbitrarily spaced grids*, Mathematics of Computation, 51 (1988), pp. 699–706.
- [8] ANDREAS GRIEWANK, JEAN UTKE, AND ANDREA WALTHER, *Evaluating higher derivative tensors by forward propagation of univariate Taylor series*, Mathematics of Computation, 69 (2000), pp. 1117–1130.
- [9] ANDREAS GRIEWANK AND ANDREA WALTHER, *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2nd ed., 2008.

- [10] DI HE, SHANDA LI, WENLEI SHI, XIAOTIAN GAO, JIA ZHANG, JIANG BIAN, LIWEI WANG, AND TIE-YAN LIU, *Learning physics-informed neural networks without stacked back-propagation*, in Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, vol. 206 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 3034–3047.
- [11] ZHEYUAN HU, ZEKUN SHI, GEORGE EM KARNIADAKIS, AND KENJI KAWAGUCHI, *Hutchinson trace estimation for high-dimensional and high-order physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering, 424 (2024), p. 116883.
- [12] ———, *Bias-variance trade-off in physics-informed neural networks with randomized smoothing for high-dimensional PDEs*, Journal of Computational Physics, 521 (2025), p. 113524.
- [13] ZHEYUAN HU, KHEMRAJ SHUKLA, GEORGE EM KARNIADAKIS, AND KENJI KAWAGUCHI, *Tackling the curse of dimensionality with physics-informed neural networks*, Neural Networks, 176 (2024), p. 106369.
- [14] MICHAEL F. HUTCHINSON, *A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines*, Communications in Statistics: Simulation and Computation, 18 (1989), pp. 1059–1076.
- [15] GEORGE EM KARNIADAKIS, IOANNIS G. KEVREKIDIS, LU LU, PARIS PERDIKARIS, SIFAN WANG, AND LIU YANG, *Physics-informed machine learning*, Nature Reviews Physics, 3 (2021), pp. 422–440.
- [16] DIEDERIK P. KINGMA AND JIMMY BA, *Adam: A method for stochastic optimization*, in International Conference on Learning Representations (ICLR), 2015.
- [17] ISAAC E. LAGARIS, ARISTIDIS LIKAS, AND DIMITRIOS I. FOTIADIS, *Artificial neural networks for solving ordinary and partial differential equations*, IEEE Transactions on Neural Networks, 9 (1998), pp. 987–1000.
- [18] RUICHEN LI, HAOTIAN YE, DU JIANG, XUELAN WEN, CHUWEI WANG, ZHE LI, XIANG LI, DI HE, JI CHEN, WEILUO REN, AND LIWEI WANG, *A computational framework for neural network-based variational Monte Carlo with Forward Laplacian*, Nature Machine Intelligence, 6 (2024), pp. 209–219.
- [19] LYNN H. LOOMIS AND SLOMO STERNBERG, *Advanced Calculus*, Jones and Bartlett, Boston, MA, revised ed., 1990.
- [20] ADAM PASZKE, SAM GROSS, FRANCISCO MASSA, ADAM LERER, JAMES BRADBURY, GREGORY CHANAN, TREVOR KILLEEN, ZEMING LIN, NATALIA GIMELSHEIN, LUCA ANTIGA, ALBAN DESMAISON, ANDREAS KÖPF, EDWARD YANG, ZACHARY DEVITO, MARTIN RAISSON, ALYKHAN TEJANI, SASANK CHILAMKURTHY, BENOIT STEINER, LU FANG, JUNJIE BAI, AND SOUMITH CHINTALA, *PyTorch: An imperative style, high-performance deep learning library*, in Advances in Neural Information Processing Systems, vol. 32, 2019.
- [21] BARAK A. PEARLMUTTER, *Fast exact multiplication by the Hessian*, Neural Computation, 6 (1994), pp. 147–160.
- [22] MAZIAR RAISSI, PARIS PERDIKARIS, AND GEORGE E. KARNIADAKIS, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics, 378 (2019), pp. 686–707.
- [23] ZEKUN SHI, ZHEYUAN HU, MIN LIN, AND KENJI KAWAGUCHI, *Stochastic Taylor derivative estimator: Efficient amortization for arbitrary differential operators*, Advances in Neural Information Processing Systems (NeurIPS), 37 (2024), pp. 122316–122353.
- [24] ———, *STDE++: Polynomial-time amortization for linear differential operators*, Journal of Machine Learning Research, 27 (2026), pp. 1–50.
- [25] YAROSLAV SHITOV, *How hard is the tensor rank?*, 2016. arXiv:1611.01559.
- [26] JUSTIN SIRIGNANO AND KONSTANTINOS SPILIOPOULOS, *DGM: A deep learning algorithm for solving partial differential equations*, Journal of Computational Physics, 375 (2018), pp. 1339–1364.
- [27] CHARLES M. STEIN, *Estimation of the mean of a multivariate normal distribution*, The Annals of Statistics, 9 (1981), pp. 1135–1151.